

0v2b note

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1 Nucleon and pion scatter

It's about the Nuclear and pion system.

1.1 Pion

The pion meson has an isospin triplet π^+ , π^0 , π^-

$$O_{\pi}^{I,I_z} = \begin{cases} \pi^+ = \bar{d}\gamma_5 u & = O_{\pi}^{1,1} & = |\pi^+\rangle \\ \pi^0 = \frac{1}{\sqrt{2}}(\bar{u}\gamma_5 u - \bar{d}\gamma_5 d) & = O_{\pi}^{1,0} & = |\pi^0\rangle \\ \pi^- = \bar{u}\gamma_5 d & = O_{\pi}^{1,-1} & = |\pi^-\rangle \end{cases} \quad (1)$$

1.2 Nuclear

The Nuclear has an isospin doublet proton p and neutron n .

$$O_N^{I,I_z} = \begin{cases} p = \epsilon_{abc} P_{\pm} u_a u_b C \gamma_5 d_c & = O_N^{\frac{1}{2},\frac{1}{2}} & = |p\rangle \\ n = \epsilon_{abc} P_{\pm} d_a u_b C \gamma_5 d_c & = O_N^{\frac{1}{2},-\frac{1}{2}} & = |n\rangle \end{cases} \quad (2)$$

1.3 Nuclear and Pion

Due to the superposition principle of isospin, it can be derived using Clebsch-Gordan coefficients that the $N\pi$ system has states with total isospin 3/2 and 1/2.

$$O_{N\pi}^{\frac{3}{2},I_z} = \begin{cases} O_N^{\frac{1}{2},\frac{1}{2}} O_{\pi}^{1,1} & = |p, \pi^+\rangle & I_z = \frac{3}{2} \\ \sqrt{\frac{2}{3}} O_N^{\frac{1}{2},\frac{1}{2}} O_{\pi}^{1,0} + \sqrt{\frac{1}{3}} O_N^{\frac{1}{2},-\frac{1}{2}} O_{\pi}^{1,1} & = \sqrt{\frac{2}{3}} |p, \pi^0\rangle + \sqrt{\frac{1}{3}} |n, \pi^+\rangle & I_z = \frac{1}{2} \\ \sqrt{\frac{1}{3}} O_N^{\frac{1}{2},\frac{1}{2}} O_{\pi}^{1,-1} + \sqrt{\frac{2}{3}} O_N^{\frac{1}{2},-\frac{1}{2}} O_{\pi}^{1,0} & = \sqrt{\frac{1}{3}} |p, \pi^-\rangle + \sqrt{\frac{2}{3}} |n, \pi^0\rangle & I_z = -\frac{1}{2} \\ O_N^{\frac{1}{2},-\frac{1}{2}} O_{\pi}^{1,-1} & = |n, \pi^-\rangle & I_z = -\frac{3}{2} \end{cases} \quad (3)$$

$$O_{N\pi}^{\frac{1}{2},I_z} = \begin{cases} -\sqrt{\frac{1}{3}} O_N^{\frac{1}{2},\frac{1}{2}} O_{\pi}^{1,0} + \sqrt{\frac{2}{3}} O_N^{\frac{1}{2},-\frac{1}{2}} O_{\pi}^{1,1} & = -\sqrt{\frac{1}{3}} |p, \pi^0\rangle + \sqrt{\frac{2}{3}} |n, \pi^+\rangle & I_z = \frac{1}{2} \\ \sqrt{\frac{2}{3}} O_N^{\frac{1}{2},\frac{1}{2}} O_{\pi}^{1,-1} - \sqrt{\frac{1}{3}} O_N^{\frac{1}{2},-\frac{1}{2}} O_{\pi}^{1,0} & = \sqrt{\frac{2}{3}} |p, \pi^-\rangle - \sqrt{\frac{1}{3}} |n, \pi^0\rangle & I_z = -\frac{1}{2} \end{cases} \quad (4)$$

1.4 Wigner-Eckart theorem

For any isospin tensor operator $T_{I_{0z}}^{(I_0)}$, its matrix element can be decomposed under the SU(2) group $m_u = m_d$ as:

$$\langle I, I_z | T_{I_{0z}}^{(I_0)} | I', I'_z \rangle = \langle I' I'_z, I_0 I_{0z} | II_z \rangle \langle I || T^{(I_0)} || I' \rangle \quad (5)$$

where $\langle I || T^{(I_0)} || I' \rangle$ is the reduced matrix element, independent of I_z, I_{0z}, I'_z .

Applied to our interpolation operator $O_{I_0, I_{0z}}$, consider its matrix element with the vacuum:

$$\langle I, I_z; n | O_{I_0, I_{0z}} | 0 \rangle = \langle I_0 I_{0z}, 00 | II_z \rangle \langle I; n | O_{I_0} | 0 \rangle. \quad (6)$$

Since the vacuum is an isospin singlet ($I = 0, I_z = 0$), the Clebsch-Gordan coefficient simplifies to $\delta_{I, I_0} \delta_{I_z, I_{0z}}$ (because $0 \otimes I_0$ can only yield I_0). In fact:

$$\langle I, I_z | O_{I_0, I_{0z}} | 0 \rangle = \delta_{I, I_0} \delta_{I_z, I_{0z}} \langle I_0 || O_{I_0} || 0 \rangle. \quad (7)$$

That is, only operators with the same isospin can create a state of that isospin from the vacuum, and the creation amplitude is independent of I_z (the reduced matrix element is the same).

More specifically, for a fixed I_0 , the ratio of the vacuum matrix elements for operators with different I_{0z} is 1 (because $\delta_{I_z, I_{0z}}$ selects the corresponding component, while the reduced matrix element is the same). Thus:

$$\langle I_0, I_z; n | O_{I_0, I_z} | 0 \rangle = \langle I_0 || O_{I_0} || 0 \rangle_n, \quad (8)$$

where $\langle I_0 || O_{I_0} || 0 \rangle_n$ is the reduced matrix element for the n -th energy level, independent of I_z .

for a fixed total isospin, the different third components have the following effect properties.

1.5 contraction

So we can use the $|p, \pi^0\rangle + |n, \pi^+\rangle$ operator to calculate the correlation function.

$$\begin{aligned} C_2^I &= (\langle n, \pi^+ | + \langle p, \pi^0 |) (|p, \pi^0\rangle + |n, \pi^+\rangle) \\ &= \langle n, \pi^+ | n, \pi^+ \rangle + \langle n, \pi^+ | p, \pi^0 \rangle + \langle p, \pi^0 | p, \pi^0 \rangle + \langle p, \pi^0 | n, \pi^+ \rangle \\ &= \langle n, \pi^+ | n, \pi^+ \rangle + \langle n, \pi^+ | p, \pi_d^0 \rangle + \langle p, \pi_u^0 | p, \pi_u^0 \rangle + \langle p, \pi_d^0 | p, \pi_d^0 \rangle + \langle p, \pi^0 | n, \pi_d^+ \rangle \\ &= \langle n, \pi^+ | n, \pi^+ \rangle + \langle n, \pi^+ | p, \pi_d^0 \rangle + \langle p, \pi_u^0 | p, \pi_u^0 \rangle + \langle p, \pi_d^0 | p, \pi_d^0 \rangle + \langle n, \pi^+ | p, \pi_d^0 \rangle^\dagger \end{aligned} \quad (9)$$

$$\frac{\vec{p}_n}{2\pi/L} = (0, 0, 0), (0, 0, 1), (0, 1, 1), (1, 1, 1) \quad (10)$$

where n is corresponding to increasing momentum modes.

$$O_{N\pi}^{(I, n)}(t_f) = \frac{1}{N_R} \sum_{\vec{R} \in O_h} \sum_{\vec{x}, \vec{y}} O_{N\pi}^{I, I_z}(\vec{x}, \vec{y}, t) e^{i\vec{R}\vec{k}(\vec{x}-\vec{y})} \quad (11)$$

where $\vec{R}$ is an element of the hypercubic group O_h , including rotations about axes (90, 180, 270) and inversions.

$$\mathcal{C}_{N\pi}^{n, m}(\Delta t) = \frac{1}{2} \langle O_{N\pi}^{(n)}(t_f) \bar{O}_{N\pi}^{(m)}(t_i) \rangle \quad (12)$$

where $\Delta t = t_f - t_i$

1.6 GEVP

set $t = \Delta t$

$$C_{N\pi}^{n,m}(t)v_n(t, t_0) = \lambda_n(t, t_0)C(t_0)v_n(t, t_0) \quad (13)$$

where $t_0 = 5$

$$\langle v_m(t, t_0)|C(t)|v_n(t, t_0)\rangle = \delta_{mn}\rho_n(t), \quad \rho_n(t) = e^{-E_n t} \quad (14)$$

$$C_{N\pi}^n(t) = \lambda_n(t, t_0) \quad (15)$$

1.7 current operator

$$\begin{aligned} V_\mu^{(1,0)} &= -\frac{1}{\sqrt{2}}(\bar{u}\gamma_\mu u - \bar{d}\gamma_\mu d) \\ V_\mu^{(0,0)} &= \frac{1}{\sqrt{2}}(\bar{u}\gamma_\mu u + \bar{d}\gamma_\mu d) \\ A_\mu^{(1,1)} &= \bar{d}\gamma_\mu\gamma_5 u \\ A_\mu^{(1,0)} &= -\frac{1}{\sqrt{2}}(\bar{u}\gamma_\mu\gamma_5 u - \bar{d}\gamma_\mu\gamma_5 d) \\ A_\mu^{(1,-1)} &= -\bar{u}\gamma_\mu\gamma_5 d \end{aligned} \quad (16)$$

$$\langle u(u^T(C\gamma_5 d))|(\bar{d}\gamma_\mu\gamma_5 u)(\bar{d}\gamma_\mu\gamma_5 u)|(\bar{u}C\gamma_5 d)d\bar{d}\gamma_5 u\rangle$$

1.8 matrix element

because of the Wigner-Eckart theorem, we can also use the total isospin to extract the matrix element

$$\mathcal{V}_\mu^{(1)} = \langle O_{N\pi}^{\frac{3}{2}, \frac{1}{2}} | V_\mu^{1,0} | O_N^{\frac{1}{2}, \frac{1}{2}} \rangle \quad (17)$$

2 five point function

The low-energy charged-current weak interaction is

$$\mathcal{L}_W(x) = 2\sqrt{2}G_F V_{ud} J_L^\mu(x) \bar{e}_L(x) \gamma_\mu \nu_L(x), \quad (18)$$

with

$$J_L^\mu(x) = \bar{u}_L(x) \gamma_\mu d_L(x). \quad (19)$$

The neutrinoless double beta decay process is second order in the weak interaction, therefore the transition amplitude is obtained from

$$S^{(2)} = \frac{i^2}{2!} \int d^4x d^4y T \{ \mathcal{L}_W(x) \mathcal{L}_W(y) \}. \quad (20)$$

2.1 pipi - ee

The matrix element between the initial hadronic state $|I\rangle$ and the final state $|F, e_1, e_2\rangle$ is

$$\mathcal{A} = \langle F, e_1, e_2 | S^{(2)} | I \rangle \quad (21)$$

$$= -4G_F^2 V_{ud}^2 \int d^4x d^4y \langle F, e_1, e_2 | T \left\{ J_L^\mu(x) \bar{e}_L(x) \gamma_\mu \nu_L(x) J_L^\nu(y) \bar{e}_L(y) \gamma_\nu \nu_L(y) \right\} | I \rangle \quad (22)$$

$$= -4G_F^2 V_{ud}^2 \int d^4x d^4y \langle F | T \left\{ J_L^\mu(x) J_L^\nu(y) \right\} | I \rangle \quad (23)$$

$$\times \langle e_1 e_2 | T \left\{ \bar{e}_L(x) \gamma_\mu \nu_L(x) \bar{e}_L(y) \gamma_\nu \nu_L(y) \right\} | 0 \rangle \quad (24)$$

For a Majorana neutrino,

$$\nu = \nu^c = C \bar{\nu}^T, \quad (25)$$

and the neutrino contraction becomes

$$\langle 0 | T \{ \nu_L(x) \bar{\nu}_L^T(y) \} | 0 \rangle = S_F(x - y) C. \quad (26)$$

where C is charge conjugate operator $C = i\gamma_4\gamma_2$

Using the left-handed projectors,

$$P_L(\not{q} + m_\nu)P_L = m_\nu P_L, \quad (27)$$

the propagator reduces to the mass insertion term,

$$S_F(x - y) \rightarrow m_{\beta\beta} \int \frac{d^4q}{(2\pi)^4} \frac{e^{-iq \cdot (x-y)}}{q^2 + i\epsilon}. \quad (28)$$

For the lepton part

$$\langle e_1 e_2 | T \left\{ \bar{e}_L(x) \gamma_\mu C \bar{e}_L(y) \gamma_\nu \right\} | 0 \rangle \times S_0(x - y) = \langle e_1 e_2 | T \left\{ \bar{e}_L(x) \gamma_\mu \gamma_\nu e_L^c(y) \right\} | 0 \rangle \times S_0(x - y) \quad (29)$$

where $e_L^c(x) = C \bar{e}_L^T$

Thus the amplitude becomes

$$\begin{aligned} \mathcal{A} = & -4G_F^2 V_{ud}^2 m_{\beta\beta} \int d^4x d^4y \langle F | T \{ J_L^\mu(x) J_L^\nu(y) \} | I \rangle S_0(x - y) \\ & \times \langle e_1, e_2 | \bar{e}_L(x) \gamma_\mu \gamma_\nu e_L^c(y) | 0 \rangle, \end{aligned} \quad (30)$$

where

$$S_0(x - y) = \int \frac{d^4q}{(2\pi)^4} \frac{e^{-iq \cdot (x-y)}}{q^2 + i\epsilon} \propto \frac{1}{2E_\nu} e^{-E_\nu(t_x - t_y)}. \quad (31)$$

For the

$$e(p) = \int d^4x a u e^{-ipx} + b v e^{ipx} \quad (32)$$

The external electron matrix element is

$$\langle e_1, e_2 | \bar{e}_L(x) \gamma_\mu \gamma_\nu e_L^c(y) | 0 \rangle = \bar{u}_L(p_1, x) \gamma_\mu \gamma_\nu u_L^c(p_2, y) - (1 \leftrightarrow 2), \quad (33)$$

with

$$\bar{u}(p, x) = \bar{u}(p) e^{E_e t_x} e^{-i \vec{p} \cdot \vec{x}}. \quad (34)$$

For the hadron part. Now insert a complete set of hadronic intermediate states,

$$1 = \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \frac{|n, \mathbf{p}_n\rangle \langle n, \mathbf{p}_n|}{2E_n}, \quad (35)$$

into the hadronic correlator:

$$\langle F | T \{ J_\mu(x) J_\nu(y) \} | I \rangle = \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \frac{\langle 0 | J_\mu(x) | n \rangle \langle n | J_\nu(y) | I \rangle}{2E_n}. \quad (36)$$

Using translation invariance,

$$J_\mu(x) = e^{iPx} J_\mu(0) e^{-iPx}, \quad (37)$$

one obtains

$$\langle F | J_\mu(x) | n \rangle = e^{i(p_F - p_n) \cdot x} \langle F | J_\mu(0) | n \rangle = e^{(E_F - E_n) \cdot t_x} e^{-i(\vec{p}_F - \vec{p}_n) \cdot \vec{x}} \langle F | J_\mu(0) | n \rangle. \quad (38)$$

The x, y integrations generate momentum conservation,

$$q = p_I - p_n - p_i, \quad (39)$$

which is the neutrino momentum flowing between the two weak vertices. For the energy part

$$\int dt_x dt_y e^{-(E_e + E_n + E_\nu - E_{\pi\pi})(t_x - t_y)} = \sum_{t_x, t_y=0}^T e^{-E_\Delta(t_x - t_y)} = 2 \frac{1}{E_\Delta} (T + \frac{e^{-E_\Delta T} - 1}{E_\Delta}) \quad (40)$$

where $E_\Delta = E_n + E_{\nu,i} + E_i - E_I$ Performing the time integrations gives the energy denominator

$$\frac{1}{E_n + E_{\nu,i} + E_i - E_I} = \frac{1}{E_\Delta}, \quad (41)$$

where

$$E_{\nu,i} = |\mathbf{p}_I - \mathbf{p}_n - \mathbf{p}_i|. \quad (42)$$

Finally, the decay amplitude is

$$\begin{aligned} \mathcal{A} = & -4G_F^2 V_{ud}^2 m_{\beta\beta} \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \sum_{i=1,2} \frac{\langle 0 | J_\mu | n \rangle \langle n | J_\mu | I \rangle}{2E_n E_{\nu,i} (E_n + E_{\nu,i} + E_i - E_I)} \\ & \times \bar{u}_L(p_1) u_L^c(p_2). \end{aligned} \quad (43)$$

Defining the leptonic factor

$$T_{\text{lept}} = 4G_F^2 V_{ud}^2 m_{\beta\beta} \bar{u}_L(p_1) u_L^c(p_2), \quad (44)$$

the amplitude can be written compactly as

$$\boxed{\mathcal{A} = -T_{\text{lept}} \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \sum_{i=1,2} \frac{\langle 0 | J_\mu^L | n \rangle \langle n | J_\mu^L | I \rangle}{2E_n 2E_{\nu,i} (E_n + E_{\nu,i} + E_i - E_I)}}. \quad (45)$$

2.2 nucleon pion - proton ee

This amplitude is just hadron part different from pipi-ee, then the hadron part can be write

$$\langle F|T\{J_L^\mu(x)J_L^\nu(y)\}|I\rangle = \langle p|T\{J_L^\mu(x)J_L^\nu(y)\}|n\pi^-\rangle \quad (46)$$

Now insert a complete set of hadronic intermediate states,

$$1 = \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \frac{|n, \mathbf{p}_n\rangle \langle n, \mathbf{p}_n|}{2E_n}, \quad (47)$$

$$\langle F|T\{J_L^\mu(x)J_L^\nu(y)\}|I\rangle = \sum_n \frac{1}{V} \sum_{\mathbf{p}_n} \frac{\langle p|J_\mu(x)|n, \mathbf{p}_n\rangle \langle n, \mathbf{p}_n|J_\nu(y)|n\pi^-\rangle}{2E_n} \quad (48)$$

For the energy part

$$\int dt_x dt_y e^{-(E_e+E_n+E_\nu-E_{\pi\pi})(t_x-t_y)} e^{E_p t_x} = \sum_{t_x, t_y=0}^T e^{-E_\Delta(t_x-t_y)} \quad (49)$$

3 three point function

The three-point correlation function can be written as

$$C_3(t, \tau) = \langle F(t)|\mathcal{O}(\tau)|I(0)\rangle. \quad (50)$$

By inserting complete sets of intermediate states,

$$\mathbf{1} = \sum_m |F_m\rangle \langle F_m|, \quad \mathbf{1} = \sum_n |I_n\rangle \langle I_n|, \quad (51)$$

the spectral decomposition becomes

$$C_3(t, \tau) = \sum_{m,n} Z_m^F Z_n^I \mathcal{M}_{mn} e^{-E_m^F(t-\tau)} e^{-E_n^I \tau}, \quad (52)$$

where

$$Z_m^F = \langle 0|\chi_F|F_m\rangle, \quad Z_n^I = \langle I_n|\bar{\chi}_I|0\rangle, \quad (53)$$

and

$$\mathcal{M}_{mn} = \langle F_m|\mathcal{O}|I_n\rangle. \quad (54)$$

In the large Euclidean-time limit only the ground states survive exponentially, giving

$$C_3(t, \tau) \rightarrow Z_0^F Z_0^I \mathcal{M}_{00} e^{-E_0^F(t-\tau)} e^{-E_0^I \tau}. \quad (55)$$

two points function

$$\begin{aligned} C_2^F(t_0, t+t_0) &= \langle F(t_0+t)|F(t_0)\rangle = \sum_n |Z_n^F|^2 e^{-E_n^F t} \\ C_2^I(t_0, t+t_0) &= \langle I(t_0+t)|I(t_0)\rangle = \sum_n |Z_n^I|^2 e^{-E_n^I t} \end{aligned} \quad (56)$$

3.1 ratio

For the ground state

$$R = \frac{C_3(t_0, \tau, t)}{C_2^F(t_0, t+t_0)} \times \sqrt{\frac{C_2^I(t_0, t-\tau)C_2^F(t_0, \tau+t_0)C_2^F(t_0, t+t_0)}{C_2^F(t_0, t-\tau)C_2^I(t_0, \tau+t_0)C_2^I(t_0, t+t_0)}} \quad (57)$$

if set $t_0 = 0$:

$$R = \frac{C_3(t, \tau)}{C_2^F(t)} \times \sqrt{\frac{C_2^I(t-\tau)C_2^F(\tau)C_2^F(t)}{C_2^F(t-\tau)C_2^I(\tau)C_2^I(t)}} \quad (58)$$

However, at finite source-sink separation the excited-state contributions cannot be neglected. Keeping only the first excited states,

$$\begin{aligned} C_3(t, \tau) = & Z_0^F Z_0^I \mathcal{M}_{00} e^{-E_0^F(t-\tau)} e^{-E_0^I \tau} + Z_1^F Z_0^I \mathcal{M}_{10} e^{-E_1^F(t-\tau)} e^{-E_0^I \tau} \\ & + Z_0^F Z_1^I \mathcal{M}_{01} e^{-E_0^F(t-\tau)} e^{-E_1^I \tau} + Z_1^F Z_1^I \mathcal{M}_{11} e^{-E_1^F(t-\tau)} e^{-E_1^I \tau} + \dots \end{aligned} \quad (59)$$

Defining the energy gaps

$$\Delta_F = E_1^F - E_0^F, \quad \Delta_I = E_1^I - E_0^I, \quad (60)$$

the three-point function can be rewritten as

$$\begin{aligned} C_3(t, \tau) = & Z_0^F Z_0^I e^{-E_0^F(t-\tau)} e^{-E_0^I \tau} \left[\mathcal{M}_{00} + \frac{Z_1^F}{Z_0^F} \mathcal{M}_{10} e^{-\Delta_F(t-\tau)} \right. \\ & \left. + \frac{Z_1^I}{Z_0^I} \mathcal{M}_{01} e^{-\Delta_I \tau} + \frac{Z_1^F Z_1^I}{Z_0^F Z_0^I} \mathcal{M}_{11} e^{-\Delta_F(t-\tau)} e^{-\Delta_I \tau} + \dots \right]. \end{aligned} \quad (61)$$

Similarly, the two-point correlation functions are expanded as

$$C_2^F(t) = |Z_0^F|^2 e^{-E_0^F t} (1 + r_F e^{-\Delta_F t} + \dots), \quad (62)$$

and

$$C_2^I(t) = |Z_0^I|^2 e^{-E_0^I t} (1 + r_I e^{-\Delta_I t} + \dots), \quad (63)$$

with

$$r_F = \frac{|Z_1^F|^2}{|Z_0^F|^2}, \quad r_I = \frac{|Z_1^I|^2}{|Z_0^I|^2}. \quad (64)$$

The ratio is defined by

$$R(t, \tau) = \frac{C_3(t, \tau)}{C_2^F(t)} \sqrt{\frac{C_2^I(t-\tau)C_2^F(\tau)C_2^F(t)}{C_2^F(t-\tau)C_2^I(\tau)C_2^I(t)}}. \quad (65)$$

After substituting the spectral decompositions of both the two-point and three-point functions, the overlap factors and the leading exponential time dependences cancel. The ratio therefore becomes

$$R(t, \tau) = \mathcal{M}_{00} + A e^{-\Delta_F(t-\tau)} + B e^{-\Delta_I \tau} + C e^{-\Delta_F(t-\tau)} e^{-\Delta_I \tau} + \dots \quad (66)$$

The first term corresponds to the desired ground-state matrix element,

$$\mathcal{M}_{00} = \langle F_0 | \mathcal{O} | I_0 \rangle, \quad (67)$$

while the remaining terms represent excited-state contaminations. The term proportional to

$$e^{-\Delta_F(t-\tau)} \quad (68)$$

comes from excited states propagating between the operator insertion and the sink. The term proportional to

$$e^{-\Delta_I \tau} \quad (69)$$

originates from excited states propagating between the source and the operator insertion. The mixed term

$$e^{-\Delta_F(t-\tau)} e^{-\Delta_I \tau} \quad (70)$$

describes simultaneous excited-state contributions from both source and sink sides.

In the asymptotic limit

$$\tau \gg 1, \quad t - \tau \gg 1, \quad (71)$$

all excited-state contributions are exponentially suppressed and the ratio approaches the desired matrix element,

$$\lim_{\tau \gg 1, t-\tau \gg 1} R(t, \tau) = \langle F_0 | \mathcal{O} | I_0 \rangle. \quad (72)$$

Therefore, in practical lattice calculations the leading systematic uncertainty at finite source-sink separation is controlled by the exponential suppression factors

$$e^{-\Delta_F(t-\tau)}, \quad e^{-\Delta_I \tau}, \quad e^{-\Delta_F(t-\tau) - \Delta_I \tau}. \quad (73)$$